FILTER DESIGN TEST CASE

High-Pass Filter - Band 7 (2600 MHz) for Satellite L-Band Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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Design Center:	Irvine Design Center
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1. OBJECTIVE

This document describes the design, simulation, and characterization of a High-Pass Filter filter targeting Band 7 (2600 MHz) for Satellite L-Band Filter. The filter is designed on AlN on SiC substrate with a target center frequency of 3959.4 MHz and bandwidth of 338.9 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Satellite L-Band Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: High-Pass Filter
- Frequency Band: Band 7 (2600 MHz)
- Application: Satellite L-Band Filter
- Substrate: AlN on SiC
- Package: Fan-Out WLP
- Design Tool: PathWave EM Design
- Design Center: Irvine Design Center

This specification applies to all variants of the High-Pass Filter design targeting Band 7 (2600 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency: 3959.4 MHz
Bandwidth (3 dB): 338.9 MHz
Insertion Loss Target: <= 3.2 dB
Return Loss Target: >= 18.0 dB
Out-of-Band Rejection: >= 43.0 dB
Isolation (Duplexer): >= 40.0 dB
Q Factor: 4014
Temperature Coefficient:-14.7 ppm/°C
Die Size: 1.6 x 0.6 mm
Wafer Size: 8-inch
Number of Resonators: 11
Electrode Material: Pt/Al

Passivation: Si3N4

4. TEST PROCEDURE

1. Open PathWave EM Design and load the High-Pass Filter template project.
2. Define the substrate stack: AlN on SiC with specified layer thicknesses.
3. Set design targets: center freq 3959.4 MHz, BW 338.9 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 3.2 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (Fan-Out WLP).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, AlN on SiC).
10. Perform on-wafer probing using Anritsu MS46122B ShockLine.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 11878 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 3.2 dB
- Return loss in passband shall be >= 18.0 dB
- Out-of-band rejection at +/- 508 MHz offset >= 43.0 dB
- Group delay variation across passband shall be <= 3 ns
- Temperature drift over -40°C to +85°C shall be <= 7.3 MHz
- Power handling shall be >= +26 dBm at 1 dB compression
- ESD tolerance >= 500 V (HBM)
- Die yield on qualification lot >= 96%

6. TEST RESULTS

Measurement Date: 2024-01-18
Devices Tested: 85
Insertion Loss (meas): 2.95 dB
Return Loss (meas): 20.5 dB
Rejection (meas): 46.3 dB
Isolation (meas): 45.0 dB
Q Factor (meas): 4014
Group Delay Variation: 3.6 ns
Power Handling: +34 dBm
Die Yield: 98.1%

Result Classification: PASS

7. OBSERVATIONS AND FINDINGS

a) The High-Pass Filter design demonstrated excellent performance for Band 7 (2600 MHz).
b) Insertion loss is marginally above target; electrode thickness optimization recommended.
c) Out-of-band rejection exceeded the minimum requirement.
d) Package parasitic effects were consistent with 3D EM model predictions.
e) Batch-to-batch reproducibility confirmed over 3 wafer lots.

8. CORRECTIVE ACTIONS

No corrective actions required. Design passed all acceptance criteria.

9. SIGN-OFF

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Approver:	Dr. Hans Weber	Date: 2024-01-28

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